

Charles Pinter - A Book of Set Theory

Chapter 1

Exercises 1.1

1. *Proof.* For all sentences P, Q and R , we are going to proof the following statments to be true, using its associated truth table. First of all, notice the truth table for the expression $P \iff Q$ (that is, the abbreviaton of $(P \Rightarrow Q) \wedge (P \Rightarrow P)$):

P	Q	$P \Rightarrow Q$	$Q \Rightarrow P$	$P \iff Q : (P \Rightarrow Q) \wedge (P \Rightarrow P)$
1	1	1	1	1
1	0	1	0	0
0	1	0	1	0
0	0	1	1	1

- i. $P \vee Q \iff Q \vee P$

P	Q	$P \vee Q$	$Q \vee P$	$P \vee Q \iff Q \vee P$
1	1	1	1	1
1	0	1	1	1
0	1	1	1	1
0	0	0	0	1

- ii. $P \vee (Q \vee R) \iff (P \vee Q) \vee R$

P	Q	R	$Q \vee R$	$P \vee Q$	$P \vee (Q \vee R)$	$(P \vee Q) \vee R$	$P \vee (Q \vee R) \iff (P \vee Q) \vee R$
1	1	1	1	1	1	1	1
1	1	0	1	1	1	1	1
1	0	1	1	1	1	1	1
1	0	0	0	1	1	1	1
0	1	1	1	1	1	1	1
0	1	0	1	1	1	1	1
0	0	1	1	0	1	1	1
0	0	0	0	0	0	0	1

- iii. $P \wedge (Q \vee R) \iff (P \wedge Q) \vee (P \wedge R)$.

Let $X = P \wedge (Q \vee R)$ and $Y = (P \wedge Q) \vee (P \wedge R)$.

P	Q	R	$Q \vee R$	$P \vee Q$	$P \wedge (Q \vee R)$	$P \wedge Q$	$P \wedge R$	$(P \wedge Q) \vee (P \wedge R)$	$X \iff Y$
1	1	1	1	1	1	1	1	1	1
1	1	0	1	1	1	1	0	1	1
1	0	1	1	1	1	0	1	1	1
1	0	0	0	0	0	0	0	0	1
0	1	1	1	0	0	0	0	0	1
0	1	0	1	0	0	0	0	0	1
0	0	1	1	0	0	0	0	0	1
0	0	0	0	0	0	0	0	0	1

- iv. $P \vee P \iff P$

P	P	$P \vee P$	$P \vee P \iff P$
1	1	1	1
0	0	0	1

□

2. *Proof.* Let's proof DeMorgan's laws using its derive truth table:

a) $\neg(P \vee Q) \iff \neg P \wedge \neg Q$

P	Q	$P \vee Q$	$\neg P$	$\neg Q$	$\neg(P \vee Q)$	$\neg P \wedge \neg Q$	$\neg(P \vee Q) \iff \neg P \wedge \neg Q$
1	1	1	0	0	0	0	1
1	0	1	0	1	0	0	1
0	1	1	1	0	0	0	1
0	0	0	1	1	1	1	1

b) $\neg(P \wedge Q) \iff \neg P \vee \neg Q$

P	Q	$P \wedge Q$	$\neg P$	$\neg Q$	$\neg(P \wedge Q)$	$\neg P \vee \neg Q$	$\neg(P \wedge Q) \iff \neg P \vee \neg Q$
1	1	1	0	0	0	0	1
1	0	0	0	1	1	1	1
0	1	0	1	0	1	1	1
0	0	0	1	1	1	1	1

□

3. *Proof.* a) $\neg\neg P \Rightarrow P$

P	$\neg P$	$\neg\neg P$	$\neg\neg P \Rightarrow P$
1	0	1	1
0	1	0	1

b) $P \Rightarrow \neg\neg P$

P	$\neg P$	$\neg\neg P$	$P \Rightarrow \neg\neg P$
1	0	1	1
0	1	0	1

□

4. *Proof.* a) $(P \Rightarrow Q) \iff \neg Q \Rightarrow \neg P$

P	Q	$\neg Q$	$\neg P$	$P \Rightarrow Q$	$\neg Q \Rightarrow \neg P$	$(P \Rightarrow Q) \iff \neg Q \Rightarrow \neg P$
1	1	0	0	1	1	1
1	0	1	0	0	0	1
0	1	0	1	1	1	1
0	0	1	1	1	1	1

b) $(P \Rightarrow Q) \iff (\neg P \vee Q)$

P	Q	$\neg P$	$P \Rightarrow Q$	$\neg P \vee Q$	$(P \Rightarrow Q) \iff \neg(P \wedge \neg Q)$
1	1	0	1	1	1
1	0	0	0	0	1
0	1	1	1	1	1
0	0	1	1	1	1

c) $(P \Rightarrow Q) \iff \neg(P \wedge \neg Q)$

P	Q	$\neg Q$	$P \Rightarrow Q$	$P \wedge \neg Q$	$\neg(P \wedge \neg Q)$	$(P \Rightarrow Q) \iff \neg(P \wedge \neg Q)$
1	1	0	1	0	1	1
1	0	1	0	1	0	1
0	1	0	1	0	1	1
0	0	1	1	0	1	1

d) $(P \wedge (P \Rightarrow Q)) \Rightarrow Q$

P	Q	$P \Rightarrow Q$	$P \wedge (P \Rightarrow Q)$	$(P \wedge (P \Rightarrow Q)) \Rightarrow Q$
1	1	1	1	1
1	0	0	0	1
0	1	1	0	1
0	0	1	0	1

e) $((P \vee Q) \wedge \neg P) \Rightarrow Q$

P	Q	$\neg P$	$P \vee Q$	$((P \vee Q) \wedge \neg P)$	$((P \vee Q) \wedge \neg P) \Rightarrow Q$
1	1	1	1	1	1
1	0	0	0	0	1
0	1	1	0	0	1
0	0	1	0	0	1

□

5. Proof. a) $((P \Rightarrow Q) \wedge (Q \Rightarrow R)) \Rightarrow (P \Rightarrow R)$

Let $X = (P \Rightarrow Q) \wedge (Q \Rightarrow R)$

P	Q	R	$P \Rightarrow Q$	$Q \Rightarrow R$	$P \Rightarrow R$	$(P \Rightarrow Q) \wedge (Q \Rightarrow R)$	$X \Rightarrow (P \Rightarrow R)$
1	1	1	1	1	1	1	1
1	1	0	1	0	0	0	1
1	0	1	0	1	1	0	1
1	0	0	0	1	0	0	1
0	1	1	1	1	1	1	1
0	1	0	1	0	1	0	1
0	0	1	1	1	1	0	1
0	0	0	1	1	1	0	1

b) $((P \Rightarrow Q) \wedge (R \Rightarrow Q)) \iff ((P \vee R) \Rightarrow Q)$

Let $X = ((P \Rightarrow Q) \wedge (R \Rightarrow Q))$

P	Q	R	$P \Rightarrow Q$	$R \Rightarrow Q$	X	$P \vee R$	$((P \vee R) \Rightarrow Q)$	$X \iff ((P \vee R) \Rightarrow Q)$
1	1	1	1	1	1	1	1	1
1	1	0	1	1	1	1	1	1
1	0	1	0	0	0	1	0	1
1	0	0	0	1	0	1	0	1
0	1	1	1	1	1	1	1	1
0	1	0	1	1	1	0	1	1
0	0	1	1	0	0	1	0	1
0	0	0	1	1	1	0	1	1

c) $((P \Rightarrow Q) \wedge (P \Rightarrow R)) \iff (P \Rightarrow (Q \wedge R))$

Let $X = ((P \Rightarrow Q) \wedge (P \Rightarrow R))$

P	Q	R	$P \Rightarrow Q$	$P \Rightarrow R$	X	$Q \wedge R$	$P \Rightarrow (Q \wedge R)$	$X \Longleftrightarrow (P \Rightarrow (Q \wedge R))$
1	1	1	1	1	1	1	1	1
1	1	0	1	0	0	0	0	1
1	0	1	0	1	0	0	0	1
1	0	0	0	0	0	0	0	1
0	1	1	1	1	1	1	1	1
0	1	0	1	1	1	0	1	1
0	0	1	1	1	1	0	1	1
0	0	0	1	1	1	0	1	1

□

6. *Proof.* a) $P \vee Q \Longleftrightarrow P \vee R$

Suppose that for all sentences Q, R and $P, Q \Longleftrightarrow R$ is true. Since $Q \Longleftrightarrow R$ is the abbreviation of $(Q \Rightarrow R) \wedge (R \Rightarrow Q)$, we can assume that $R \Rightarrow Q$ is true. Then, let's derive the truth table for $P \vee R \Rightarrow P \vee Q$

P	Q	R	$P \vee R$	$P \vee Q$	$((P \vee R) \Rightarrow (P \vee Q))$
1	1	1	1	1	1
1	1	0	1	1	1
1	0	1	1	1	1
1	0	0	1	1	1
0	1	1	1	1	1
0	1	0	0	1	1
0	0	1	1	0	0
0	0	0	0	0	1

In this manner, since $R \Rightarrow Q$ is true, we cannot have simultaneously, R true and Q false, then we may disregard the seventh line of the table. In all the remaining lines, $P \vee Q \Rightarrow P \vee R$ takes the value 1 (true). Finally, because $Q \Rightarrow R$ is also true, Theorem 1.7 also holds, then $P \vee Q \Rightarrow Q \vee R$ and $P \vee R \Rightarrow Q \vee R$ are true, that is $P \vee Q \Longleftrightarrow P \vee R$ is true.

b) Analogous to the above.

c) $(P \Rightarrow Q) \Longleftrightarrow (P \Rightarrow R)$.

Suppose $Q \Longleftrightarrow R$ is true. Let derive the truth table for $(P \Rightarrow Q) \Longleftrightarrow (P \Rightarrow R)$

P	Q	R	$P \Rightarrow Q$	$P \Rightarrow R$	$(P \Rightarrow Q) \Longleftrightarrow (P \Rightarrow R)$
1	1	1	1	1	1
1	1	0	0	0	0
1	0	1	0	1	0
1	0	0	1	0	1
0	1	1	1	1	1
0	1	0	1	1	1
0	0	1	1	1	1
0	0	0	1	1	1

Since $Q \Rightarrow R$ and $R \Rightarrow Q$ is true, we cannot have simultaneously Q true and R false, and P true and R false. Then, we may disregard the second and third line of the table. Finally, in all the remaining lines $(P \Rightarrow Q) \Longleftrightarrow (P \Rightarrow R)$ takes the value 1 (true).

□

7. *Proof.* Suppose $P \Rightarrow Q$ and $R \Rightarrow S$ are true.

a) $P \vee R \Rightarrow Q \vee S$ is true.

Since $P \Rightarrow Q$ is true, for Theorem 1.7 we have that

$$P \vee R \iff R \vee P \Rightarrow R \vee Q \iff Q \vee R$$

and, since $R \Rightarrow S$ is true, then $Q \vee R \Rightarrow Q \vee S$ is true. Therefore, for exercise 5. (a) we have that

$$((P \vee R \Rightarrow Q \vee R) \wedge (Q \vee R \Rightarrow Q \vee S)) \Rightarrow (P \vee R \Rightarrow Q \vee S)$$

is true for all P, Q, R and S . Then, we cannot have simultaneously $((P \vee R \Rightarrow Q \vee R) \wedge (Q \vee R \Rightarrow Q \vee S))$ true and $(P \vee R \Rightarrow Q \vee S)$ false. But $((P \vee R \Rightarrow Q \vee R) \wedge (Q \vee R \Rightarrow Q \vee S))$ is always true, therefore $(P \vee R \Rightarrow Q \vee S)$ has to be true.

b) Analogous to the previous one, using Theorem 1.7 (ii).

□

Charles Pinter - A Book of Set Theory (Formal Methods)

Chapter 1

Exercises 1.1

Library chapter_1_exercises_1_point_1

This is basically checking for every possibility. It simply splits the directions and the options and then applies them one by one.

Theorem *theorem_1_8_i* : $\forall (P \ Q: \text{Prop}), P \vee Q \leftrightarrow Q \vee P$.

Proof.

```
intros P Q. split; intros [H1 | H2].
  ** right. apply H1.
  ** left. apply H2.
  ** right. apply H1.
  ** left. apply H2.
```

Qed.

Similar to the last point, it splits the possibilities and applies them as needed.

Theorem *theorem_1_8_ii* : $\forall (P \ Q \ R: \text{Prop}),$

$P \vee (Q \vee R) \leftrightarrow (P \vee Q) \vee R$.

Proof.

```
intros P Q R. split.
× intros [HP | [HQ | HR]].
  ** left. left. apply HP.
  ** left. right. apply HQ.
  ** right. apply HR.
× intros [[HP | HQ] | HR].
  ** left. apply HP.
  ** right. left. apply HQ.
  ** right. right. apply HR.
```

Qed.

You will become accustomed to the way it works. Simply splitting the possibilities and applying them is basically what is done here.

Theorem *theorem_1-8-iii* : $\forall (P \ Q \ R : \text{Prop}), P \wedge (Q \vee R) \leftrightarrow (P \wedge Q) \vee (P \wedge R)$.

Proof.

```

intros P Q R. split.
× intros [HP [HQ | HR]].
  ** left. split. apply HP. apply HQ.
  ** right. split. apply HP. apply HR.
× intros [[HP HQ] | [HP HR]].
  ** split. apply HP. left. apply HQ.
  ** split. apply HP. right. apply HR.

```

Qed.

One important note in this exercise (one that I strongly recommend looking into) is the use of `;` in this proof. When you use it after opening different branches, it applies the command in both branches (really cool for solving many problems at the same time).

Theorem *theorem_1-8-iv* : $\forall (P : \text{Prop}), P \vee P \leftrightarrow P$.

Proof.

```

intros P. split.
× intros [HP | HP]; apply HP.
× intros HP; left; apply HP.

```

Qed.

As before, it is simply applying every option. Note that it is simply reversing the order of application.

Theorem *theorem_1-8-i'* : $\forall (P \ Q : \text{Prop}), P \wedge Q \leftrightarrow Q \wedge P$.

Proof.

```

intros P Q. split.
× intros [HP HQ]. split. apply HQ. apply HP.
× intros [HQ HP]. split. apply HP. apply HQ.

```

Qed.

You can (and I actually did) try to simplify this proof with `;` and try, however it is uglier and more convoluted that way.

Theorem *theorem_1-8-ii'* : $\forall (P \ Q \ R : \text{Prop}), P \wedge (Q \wedge R) \leftrightarrow (P \wedge Q) \wedge R$.

Proof.

```

intros P Q R. split.
× intros [HP [HQ HR]]. split. split. apply HP. apply HQ. apply HR.
× intros [[HP HQ] HR]. split. apply HP. split. apply HQ. apply HR.

```

Qed.

It is in the same direction as every other proof in this first exercise. Simply working every possibility.

Theorem *theorem_1-8-iii'* : $\forall (P \ Q \ R : \text{Prop}), P \vee (Q \wedge R) \leftrightarrow (P \vee Q) \wedge (P \vee R)$.

Proof.

```

intros P Q R. split.
× intros [HP | [HQ HR]].
  ** split; left; apply HP.
  ** split; right. apply HQ. apply HR.
× intros [[HP1 | HQ] [HP2 | HR]].
  ** left. apply HP1.
  ** left. apply HP1.
  ** left. apply HP2.
  ** right. split. apply HQ. apply HR.

```

Qed.

We are ignoring one of the hypotheses because it doesn't add anything to have two hypotheses which both simply state that P is true.

Theorem *theorem_1_8_iv'* : $\forall (P : \text{Prop}), P \wedge P \leftrightarrow P$.

Proof.

```
  intros P. split.
  × intros [HP _].
    apply HP.
  × intros HP. split; apply HP.
Qed.
```

Library chapter_1_exercises_1_point_2

From *OUMBS* Require Import *chapter_1_exercises_1_point_1*.

Theorem *exercise_1_2_a* : $\forall (P\ Q : \text{Prop}), \sim(P \vee Q) \leftrightarrow (\sim P) \wedge (\sim Q)$.

Proof.

```
  intros P Q.
  unfold not. split.
  × intros HPQ. split.
    ** intros HP. apply or_introl with (B := Q) in HP. apply HPQ in HP. apply HP.
    ** intros HQ. apply or_introl with (B := P) in HQ. apply theorem_1_8_i in HQ. apply HPQ in
    HQ. apply HQ.
  × intros [HNP HNQ]. intros [HP | HQ]. apply HNP in HP. apply HP. apply HNQ in HQ. apply
  HQ.
Qed.
```

Axiom *invalid_middle* : $\forall (P : \text{Prop}), P \vee \neg P$.

Theorem *exercise_1_2_b* : $\forall (P\ Q : \text{Prop}), \sim(P \wedge Q) \leftrightarrow (\sim P) \vee (\sim Q)$.

Proof.

```
  intros P Q.
  unfold not. split.
  × intros HNPQ. destruct (invalid_middle P) as [HP | HNP].
    ** right. intros HQ. assert (H : P ∧ Q). { split. apply HP. apply HQ. }
    apply HNPQ in H. apply H.
    ** left. apply HNP.
  × intros [HNP | HNQ].
    ** intros HPQ. apply proj1 in HPQ. apply HNP in HPQ. apply HPQ.
    ** intros HPQ. apply proj2 in HPQ. apply HNQ in HPQ. apply HPQ.
Qed.
```

Library chapter_1_exercises_1_point_3

From *OUMBS* Require Import *chapter_1_exercises_1_point_2*.

Theorem *exercise_1_3_a* : $\forall (P : \text{Prop}), \sim\sim P \leftrightarrow P$.

Proof. split.

unfold not.

```
× intros H. destruct (invalid_middle P) as [HP | HNP]. apply HP. apply H in HNP. destruct HNP.
× intros HP. unfold not. intros HNP. apply HNP in HP. apply HP.
Qed.
```

Library chapter_1_exercises_1_point_4

From *OUMBS* Require Import *chapter_1_exercises_1_point_2*.

Theorem *exercise_1_4_a* : $\forall (P\ Q : \text{Prop}), (P \rightarrow Q) \leftrightarrow (\sim Q \rightarrow \neg P)$.

Proof.

```
  intros P Q. split.
  × intros HPQ. unfold not. intros HNQ. intros HP. apply HPQ in HP. apply HNQ in HP. apply
HP.
  × unfold not. intros HNPNQ. intros HP. destruct (invalid_middle Q).
  ** apply H.
  ** apply HNPNQ in H. destruct H. apply HP.
```

Qed.

Theorem *exercise_1_4_b* : $\forall (P\ Q : \text{Prop}), (P \rightarrow Q) \leftrightarrow (\sim P \vee Q)$.

Proof.

```
  intros P Q. split.
  × intros HPQ. destruct (invalid_middle P). apply HPQ in H. right. apply H.
  left. apply H.
  × unfold not. intros [HNP | HQ].
  ** intros HP. apply HNP in HP. destruct HP.
  ** intros HP. apply HQ.
```

Qed.

Theorem *exercise_1_4_c* : $\forall (P\ Q : \text{Prop}), (P \rightarrow Q) \leftrightarrow \sim(P \wedge \neg Q)$.

Proof.

```
  intros P Q. split.
  × intros HPQ. unfold not. intros [HP HNQ]. apply HPQ in HP. apply HNQ in HP. apply HP.
  × unfold not. intros HNPNQ HP. destruct (invalid_middle Q).
  ** apply H.
  ** assert (H1 : P ∧ ¬Q). { split. apply HP. apply H. } apply HNPNQ in H1. destruct H1.
```

Qed.

Theorem *exercise_1_4_d* : $\forall (P\ Q : \text{Prop}), (P \wedge (P \rightarrow Q)) \rightarrow Q$.

Proof.

```
  intros P Q [HP HPQ].
  apply HPQ in HP. apply HP.
```

Qed.

Theorem *exercise_1_4_e* : $\forall (P\ Q : \text{Prop}), ((P \vee Q) \wedge \neg P) \rightarrow Q$.

Proof.

```
  intros P Q [[HP | HQ] HNP].
  × apply HNP in HP. destruct HP.
  × apply HQ.
```

Qed.

Library chapter_1_exercises_1_point_5

From OUMBS Require Import *chapter_1_exercises_1_point_1*.

Theorem *exercise_1_5_a* : $\forall (P\ Q\ R : \text{Prop}), ((P \rightarrow Q) \wedge (Q \rightarrow R)) \rightarrow (P \rightarrow R)$.

Proof.

```
  intros P Q R [HPQ HQR] H.
  apply HPQ in H. apply HQR in H. apply H.
```

Qed.

Theorem *exercise_1_5_b* : $\forall (P\ Q\ R : \text{Prop}), ((P \rightarrow Q) \wedge (R \rightarrow Q)) \leftrightarrow ((P \vee R) \rightarrow Q)$.

Proof.

```
  intros P Q R. split.
```



```

× intros [HPQ HQR] [HP | HR].
  ** apply HPQ in HP. apply HP.
  ** apply HQR in HR. apply HR.
× intros HPRQ. split.
  ** intros HP. apply or_introl with (B := R) in HP. apply HPRQ. apply HP.
  ** intros HR. apply or_introl with (B := P) in HR. apply theorem_1_8_i in HR. apply HPRQ.
apply HR.
Qed.

Theorem exercise_1_5_c : ∀ (P Q R : Prop), ((P → Q) ∧ (P → R)) ↔ (P → (Q ∧ R)).
Proof.
  intros P Q R. split.
  × intros [HPQ HPR] HP. split.
    ** apply HPQ. apply HP.
    ** apply HPR. apply HP.
  × intros HPQR. split; intros HP; apply HPQR; apply HP.
Qed.

```

Library chapter_1_exercises_1_point_6

```

Theorem exercise_1_6_a : ∀ (P Q R : Prop), (Q ↔ R) → (P ∨ Q ↔ P ∨ R).
Proof.
  intros P Q R HQR. split.
  × intros [HP | HQ].
    ** left. apply HP.
    ** right. apply HQR. apply HQ.
  × intros [HP | HR].
    ** left. apply HP.
    ** right. apply HQR. apply HR.
Qed.

```

```

Theorem exercise_1_6_b : ∀ (P Q R : Prop), (Q ↔ R) → (P ∧ Q ↔ P ∧ R).
Proof.
  intros P Q R HQR. split.
  × intros [HP HQ]. split.
    ** apply HP.
    ** apply HQR. apply HQ.
  × intros [HP HR]. split.
    ** apply HP.
    ** apply HQR. apply HR.
Qed.

```

```

Theorem exercise_1_6_c : ∀ (P Q R : Prop), (Q ↔ R) → ((P → Q) ↔ (P → R)).
Proof.
  intros P Q R HQR. split.
  × intros HPQ HP. apply HQR. apply HPQ. apply HP.
  × intros HPR HP. apply HQR. apply HPR. apply HP.
Qed.

```

Exercises 1.2

1. a) $A \cup C \subseteq B \cup D$

Proof. First, suppose that $A \subseteq B$ and $C \subseteq D$, that is by definition that $x \in A \Rightarrow x \in B$ and $x \in C \Rightarrow x \in D$. Then, by exercise 1.1.7(a) we can say that $x \in A \vee x \in C \Rightarrow x \in B \vee x \in D$ and by definition 1.13 we got $x \in A \cup C \Rightarrow x \in B \cup D$. Finally, we can conclude $A \cup C \subseteq B \cup D$ by definition 1.10. $\square$

b) Analogous to the previous exercise, using exercise 1.1.7(b).

2. a) $A \cup C = B \cup D$

Proof. First, suppose that $A = B$ and $C = D$. We have to prove just that $B \cup D \subseteq A \cup C$ since $A \cup C \subseteq B \cup D$ was proved in the preceding exercise, having in mind that the hypothesis of the exercise is met, because we are supposing $A = B$ and $C = D$, so, we got by the axiom of extent $x \in A \Rightarrow x \in B$, $x \in B \Rightarrow x \in A$ and $x \in C \Rightarrow x \in D$, $x \in D \Rightarrow x \in C$, that is, $A \subseteq B$, $B \subseteq A$ and $C \subseteq D$, $D \subseteq C$. Then, because of $x \in B \Rightarrow x \in A$ and $x \in D \Rightarrow x \in C$, we have

$$x \in B \vee x \in D \Rightarrow x \in A \vee x \in C$$

by exercise 1.1.7(a) and by definition 1.13 we got $x \in B \cup D \Rightarrow x \in A \cup C$. Finally, we can say that $B \cup D \subseteq A \cup C$ and $A \cup C \subseteq B \cup D$ and by theorem 1.11(iv) we conclude that $A \cup C = B \cup D$. $\square$

b) Analogous to the previous exercise, using exercise 1.1.7(b).

3. *Proof.* Let's suppose $A \subseteq B$. Then we have $x \in A \Rightarrow x \in B$ by the definition 1.10 and since $(P \Rightarrow Q) \Leftrightarrow (\neg Q \Rightarrow \neg P)$ by exercise 1.1.4(a), we got $x \notin B \Rightarrow x \notin A$, that is, $x \in B' \Rightarrow x \in A'$ by definition 1.19 and finally, we can conclude by definition 1.10 that $B' \subseteq A'$. $\square$

4. *Proof.* Since $A = B$, we have that $A \subseteq B$ and $B \subseteq A$ as a consequence of the axiom of extent and then, we can apply the preceding exercise and the theorem 1.11 (iv) to conclude $A' = B'$. $\square$

5. *Proof.* Suppose $A = B$ and $B \subseteq C$. We know by the axiom of extent that $x \in A \Rightarrow x \in B$ (i) and $x \in B \Rightarrow x \in A$. Also, we have $x \in B \Rightarrow x \in C$ and, because of exercise 1.1.5(a) we can say that $x \in A \Rightarrow x \in C$ and then we can conclude $A \subseteq C$ by definition 1.10. $\square$

6. *Proof.* Suppose $A \subset B$ and $B \subset C$. We want prove that $x \in A \Rightarrow x \in C$ and $A \neq C$. By the hypothesis we have $x \in A \Rightarrow x \in B$ and $x \in B \Rightarrow x \in C$ (i), with $A \neq B \neq C$ and then, applying exercise 1.5(a) we can say that $x \in A \Rightarrow x \in C$. Now, let's suppose $A = C$. Since $C = A \subset B$, we have by definition 1.10 that $x \in C \Rightarrow x \in B$ and if we use that fact and (i), applying the axiom of extent we got that $B = C$ (absurd). Finally, we can conclude $x \in A \Rightarrow x \in C$ and $A \neq C$, that is, $A \subset C$. $\square$

7. iv) *Proof.* Since $A \subseteq B$ and $B \subseteq A$ is by definition 1.10 equivalent to say $x \in A \Rightarrow x \in B$ and $x \in B \Rightarrow x \in A$, we have by the axiom of extent that $A = B$. $\square$

v) Just use the definition and apply exercise 1.1.5(a).

8. Suppose that $S \in S$. By definition of S we will have that $S \notin S$, which is a contradiction. Then we can conclude that S is not an element.

9. No, because A2 just permit us to form the class of all elements that satisfy a property $P(x)$, not a class of classes that satisfy the property.

10. It cannot be produced because A2 just allow to construct classes which its elements satisfy a condition, not to form a class of classes that satisfy some property.

Exercises 1.3